# 15.2: Sex-linked genes exhibit unique patterns of inheritance

Morgan's discovery of a trait (white eyes) that correlated with the sex of flies provided important support for the chromosome theory of inheritance. Because the identity of the sex chromosomes in an individual could be inferred by observing the sex of the fly, the behavior of the two members of the pair of sex chromosomes could be correlated with the behavior of the two alleles of the eye color gene. In this section, we'll take a closer look at the role of sex chromosomes in inheritance.

## The Chromosomal Basis of Sex

Although sex has traditionally been described as binary—male or female—we are coming to understand that this classification may be too simplistic. Here, we use the term *sex* to refer to classification into a group with a shared set of anatomical and physiological traits. In this sense, sex in many species is determined largely by inheritance of sex chromosomes.

Humans and other mammals have two types of sex chromosomes, designated Xand Y. The Y chromosome is much smaller than the X chromosome **(Figure 15.5)**. A person who inherits two X chromosomes, one from each parent, usually develops anatomy we associate with the "female" sex, while "male" properties are associated with the inheritance of one X chromosome and one Y chromosome **(Figure 15.6a)**. Short segments at either end of the Y chromosome are the only regions that are homologous with regions on the X. These homologous regions allow the X and Y chromosomes in males to pair and behave like homologs during meiosis in the testes.

**Figure 15.5**

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Human sex chromosomes (duplicated).
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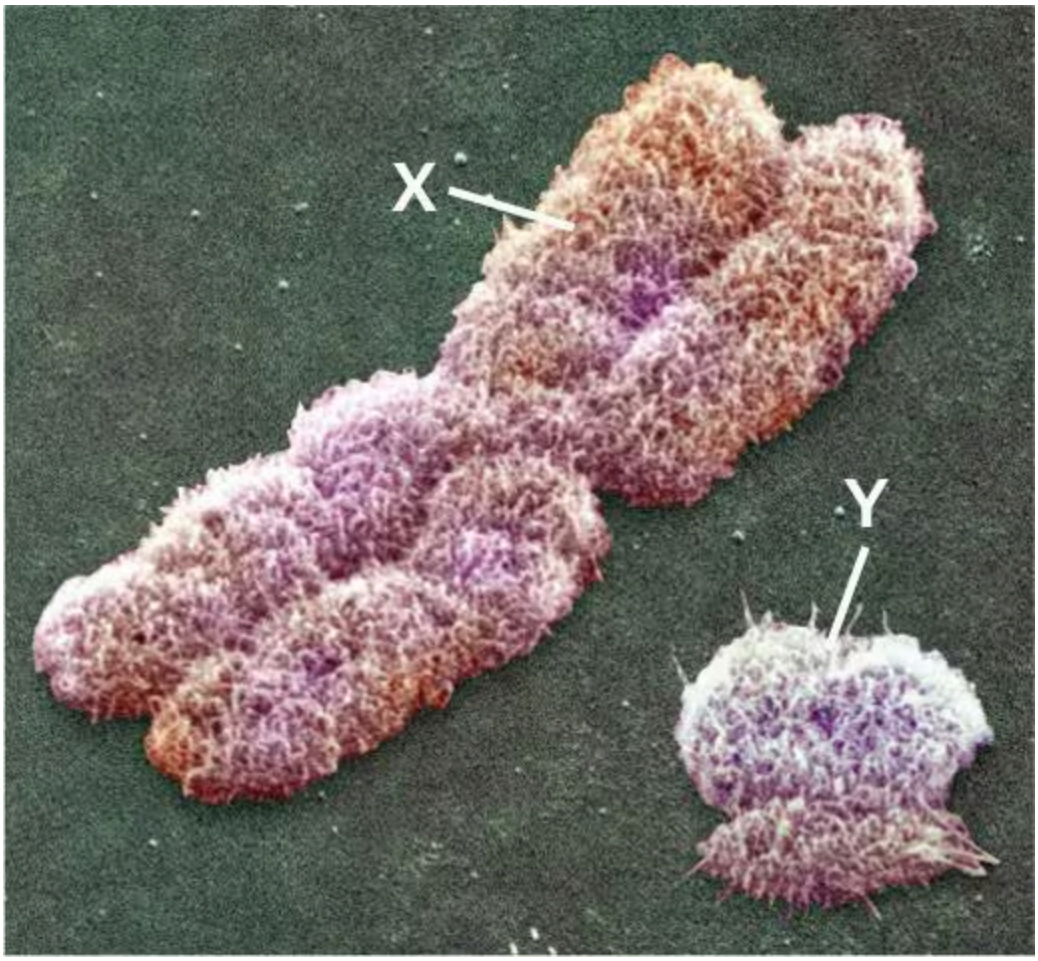
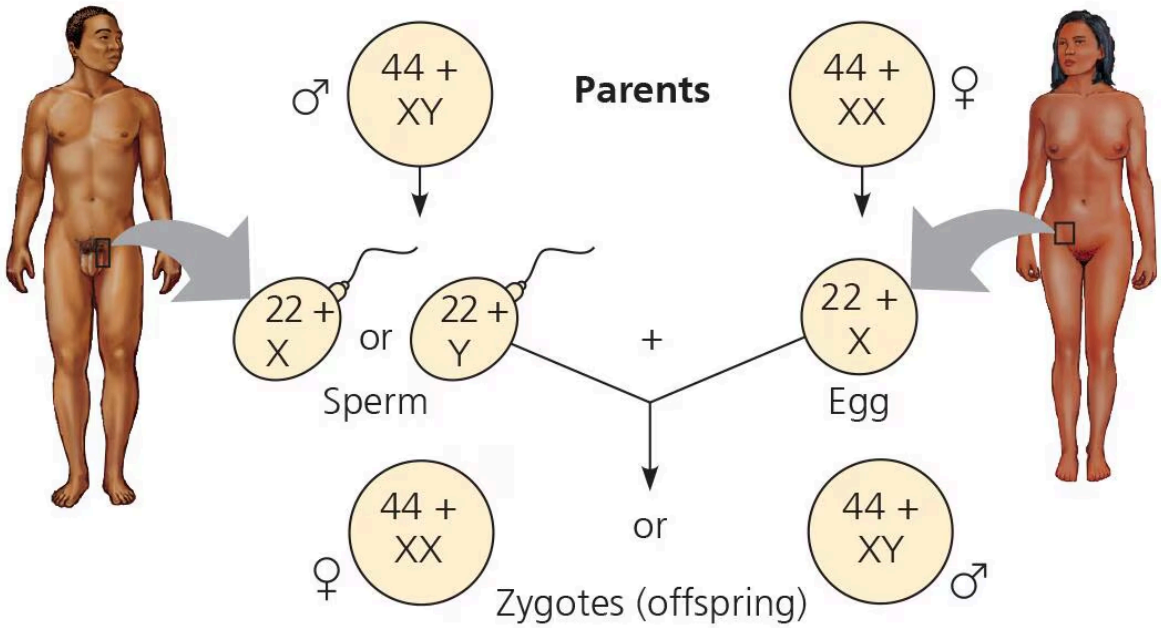
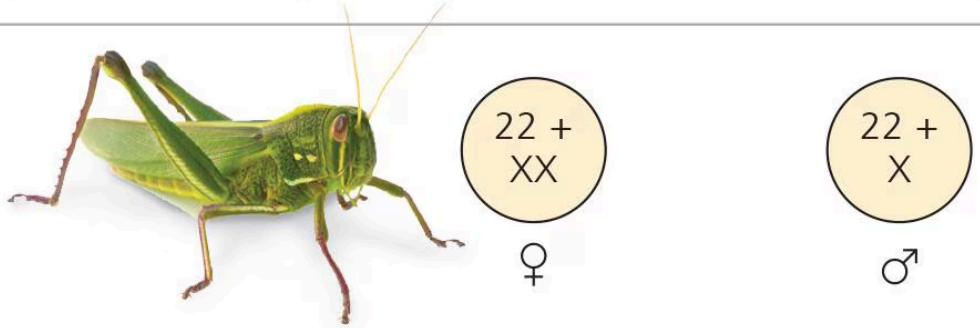



**Figure 15.6**

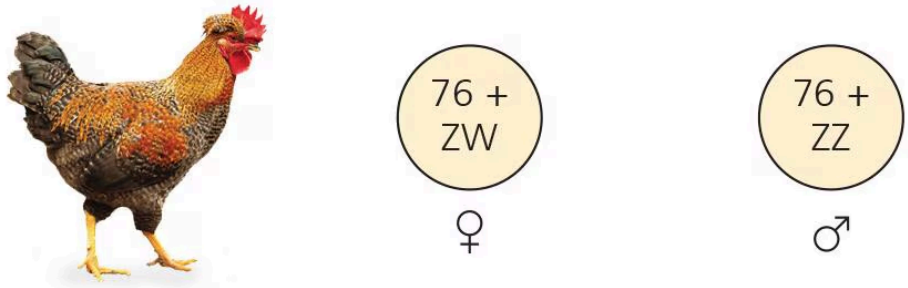
Some chromosomal systems of sex determination.




**(a) The X-Y system.** In mammals, the sex of an offspring depends on whether the sperm cell contains an X chromosome or a Y.




**(b) The X-0 system.** In grasshoppers, cockroaches, and some other insects, there is only one type of sex chromosome, the X. Females are XX; males have only one sex chromosome (X0). Sex of the offspring is determined by whether the sperm cell contains an X chromosome or no sex chromosome.




**(c) The Z-W system.** In birds, some fishes, and some insects, the sex chromosomes present in the egg (not the sperm) differ, and thus determine the sex of offspring. The sex chromosomes are designated Z and W. Females are ZW and males are ZZ.

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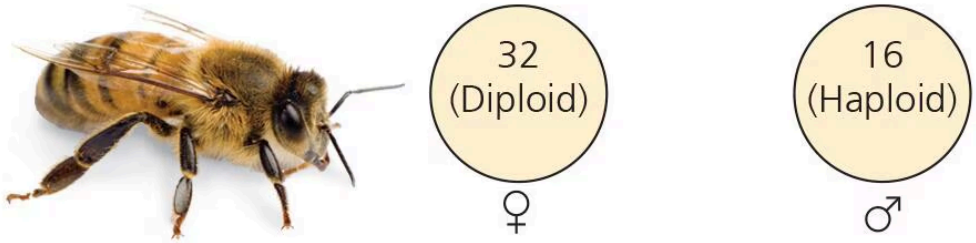



**(d) The haplo-diploid system.** There are no sex chromosomes in most species of bees and ants. Females develop from fertilized eggs and are thus diploid. Males develop from unfertilized eggs and are haploid; they have no fathers.

The systems are as follows.

X Y system. In mammals, the sex of an offspring depends on whether the sperm cell contains an X chromosome or a Y. Diagram shows sex chromosomes in the parents (X X or X Y), the gametes produced by parents (X or Y), and the sex chromosomes following fertilization in the zygotes (X X or X Y). Parents: Human male has 44 chromosomes plus an X and a Y chromosome. A Female has 44 chromosomes plus 2 X chromosomes. Sperm made by males contains 22 chromosomes plus either an X or a Y chromosome. Eggs made by females contain 22 chromosomes plus an X chromosome. After fertilization, zygotes (offspring) carry 44 chromosomes plus 2 X chromosomes if female, and 44 chromosomes plus an X and Y chromosome if male.

The X 0 system. In grasshoppers, cockroaches, and some other insects, there is only one type of sex chromosome, the X. Females are X X; males have only one sex chromosome (X 0). The sex of the offspring is determined by whether the sperm cell contains an X chromosome or no sex chromosome.

The Z W system. In birds, some fishes, and some insects, the sex chromosomes present in the egg (not the sperm) determine the sex of offspring. The sex chromosomes are designated Z and W. Females are Z W and males are Z Z.

The haplo-diploid system. There are no sex chromosomes in most species of bees and ants. Females develop from fertilized eggs and are thus diploid. Males develop from unfertilized eggs and are haploid; they have no fathers.

Numerals indicate the number of autosomes in the species pictured. In *Drosophila*, males are XY, but sex depends on the ratio of the number of X chromosomes to the number of autosome sets, not simply on the presence of a Y chromosome. In some species (not shown here), sex is determined not by chromosomes but by environmental factors such as temperature.

In mammalian testes and ovaries, the two sex chromosomes segregate during meiosis. Each egg receives one X chromosome. In contrast, sperm fall into two categories: Half the sperm cells a male produces receive an X chromosome, and half receive a Y chromosome. We can trace the sex of each offspring to the events of conception: If a sperm cell bearing an X chromosome fertilizes an egg, the zygote is XX, a female; if a sperm cell containing a Y chromosome fertilizes an egg, the zygote is XY, a male (see Figure 15.6a). Thus, in general, sex determination is a matter of chance—a fifty-fifty chance. Note that the mammalian X-Y system isn't the only chromosomal system for determining sex. **Figure 15.6b–d** illustrates three other systems.

In humans, the anatomical signs of sex begin to emerge when the embryo is about 2 months old. Before then, the rudiments of the gonads are generic—they can develop into either testes or ovaries, depending on whether or not a Y chromosome is present, and depending on what genes are active. A gene on the Y chromosome—called *SRY*, for sex-determining region of Y——is required for the development of testes. In the absence of *SRY*, the gonads develop into ovaries, even in an XY embryo.

A gene located on either sex chromosome is called a **sex-linked gene**. The human X chromosome contains approximately 1,100 genes, which are called **X-linked genes**, while genes located on the Y chromosome are called *Y-linked genes.* On the human Y chromosome, researchers have identified 78 genes that code for about 25 proteins (some genes are duplicates). About half of these genes are expressed only in the testis, and some are required for normal testicular functioning and the production of normal sperm. The Y chromosome is passed along virtually intact from a father to all his sons. Because there are so few Y-linked genes, very few disorders are transferred from father to son on the Y chromosome.

The development of female gonads in humans requires a gene called WNT4 (on chromosome 1, an autosome), which encodes a protein that promotes ovary development. An embryo that is XY but has extra copies of the WNT4 gene can develop rudimentary female gonads. Overall, sex is determined by the interactions of a network of gene products like these.

The biochemical, physiological, and anatomical features associated with males and females are turning out to be more complicated than previously thought, with many genes involved in their development. Because of the complexity of this process, many variations exist: Some individuals vary in the number of sex chromosomes in their cells (see Concept 15.4), and others are born with intermediate sexual (*intersex*) characteristics. Sex determination is an active area of research that will likely yield a more sophisticated understanding in years to come.

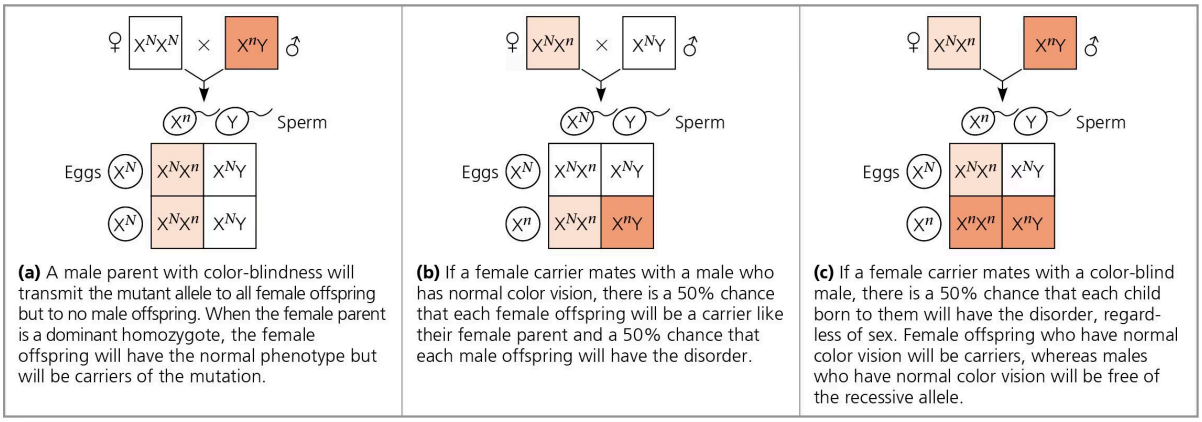
## Inheritance of X-Linked Genes

The fact that males and females inherit a different number of X chromosomes leads to a pattern of inheritance different from that produced by genes located on autosomes. While there are very few Y-linked genes, many of which help determine sex, the X chromosomes have numerous genes for characters unrelated

to sex. X-linked genes in humans follow the same pattern of inheritance that Morgan observed for the eye color locus he studied in *Drosophila* (see Figure 15.3). Fathers pass X-linked alleles to all of their daughters but to none of their sons. In contrast, mothers can pass X-linked alleles to both sons and daughters, as shown in **Figure 15.7** for the inheritance of a mild X-linked disorder, red-green color blindness.

**Figure 15.7**

The transmission of X-linked recessive traits.




**(a)** A male parent with color-blindness will transmit the mutant allele to all female offspring but to no male offspring. When the female parent is a dominant homozygote, the female offspring will have the normal phenotype but will be carriers of the mutation.

**(b)** If a female carrier mates with a male who has normal color vision, there is a 50% chance that each female offspring will be a carrier like their female parent and a 50% chance that each male offspring will have the disorder.

**(c)** If a female carrier mates with a color-blind male, there is a 50% chance that each child born to them will have the disorder, regardless of sex. Female offspring who have normal color vision will be carriers, whereas males who have normal color vision will be free of the recessive allele.

Part a. A color-blind father will transmit the mutant allele to all daughters but to no sons. When the mother is a dominant homozygote, the daughters will have the normal phenotype but will be carriers of the mutation. Punnett square shows the Cross between normal female X uppercase N X uppercase N and affected male (X lowercase n Y). Eggs are both X uppercase N. Sperm are X lowercase n Y. Results of fertilization shown inside Punnett square:

- X uppercase N egg crossed with X lowercase n sperm = X uppercase N X lowercase n; carrier daughter.
- X uppercase N egg crossed with X lowercase n sperm = X uppercase N X lowercase n; carrier daughter.
- X uppercase N egg crossed with Y sperm = X uppercase N Y; unaffected son.
- X uppercase N egg crossed with Y sperm = X uppercase N Y; unaffected son.

Part b. If a carrier mates with a male who has normal color vision, there is a 50% chance that each daughter will be a carrier like her mother and a 50% chance that each son will have the disorder. Punnett square shows cross between carrier female (X uppercase N X lowercase n) and unaffected male (X uppercase N Y). Eggs are X uppercase N and X lowercase n. Sperm are X uppercase N and Y. Results of fertilization inside the Punnett square:

- X uppercase N egg crossed with X uppercase N sperm = X uppercase N X uppercase N; unaffected daughter.

- X lowercase n egg crossed with X uppercase N sperm = X uppercase N X lowercase n; carrier daughter.
- X uppercase N egg crossed with Y sperm = X uppercase NY; unaffected son.
- X lowercase n egg crossed with Y sperm = X lowercase n Y; affected son.

Part c. If a carrier mates with a color-blind male, there is a 50% chance that each child born to them will have the disorder, regardless of sex. Daughters who have normal color vision will be carriers, whereas males who have normal color vision will be free of the recessive allele. Punnett square shows Cross between carrier female (X uppercase N X lowercase n) and affected male (X lowercase n Y). Eggs are X uppercase N and X lowercase n. Sperm are X lowercase n and Y. Results of fertilization shown inside Punnett square:

- X uppercase N egg crossed with X lowercase n sperm = X uppercase N X lowercase n ; carrier daughter.
- X lowercase n egg crossed with X lowercase n sperm = X lowercase n X lowercase n; affected daughter.
- X uppercase N egg crossed with Y sperm = X uppercase N Y; unaffected son.
- X lowercase n egg crossed with Y sperm = X lowercase n Y; affected son.

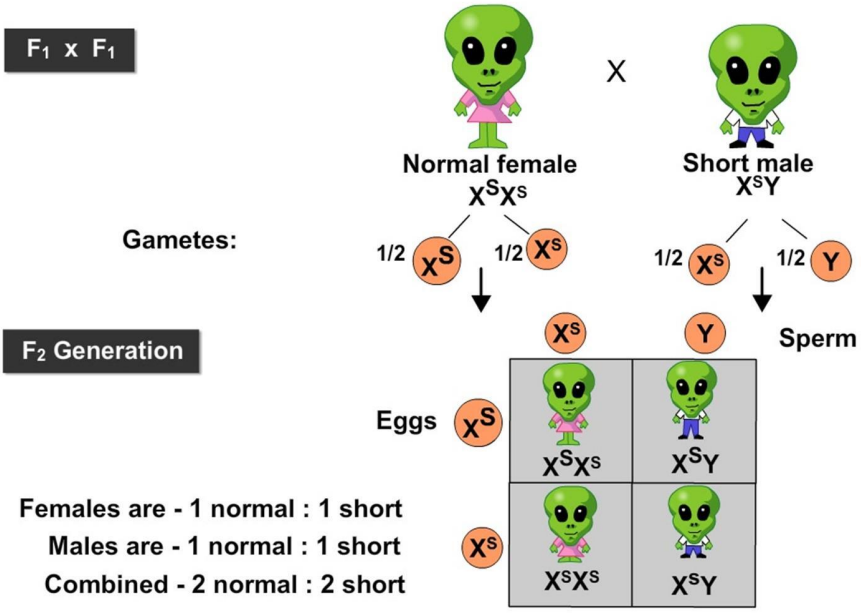
In this diagram, red-green color blindness is used as an example. The superscript *N* represents the dominant allele for normal color vision carried on the X chromosome, while *n* represents the recessive allele, which has a mutation causing color blindness. White boxes indicate unaffected individuals, light orange boxes indicate carriers, and dark orange boxes indicate color-blind individuals.

*If a color-blind female married a male who had normal color vision, what would be the probable phenotypes of their children?*

*For suggested answer, see*Appendix A*.*

**Animation**

X-Linked Genes in “MendAliens”

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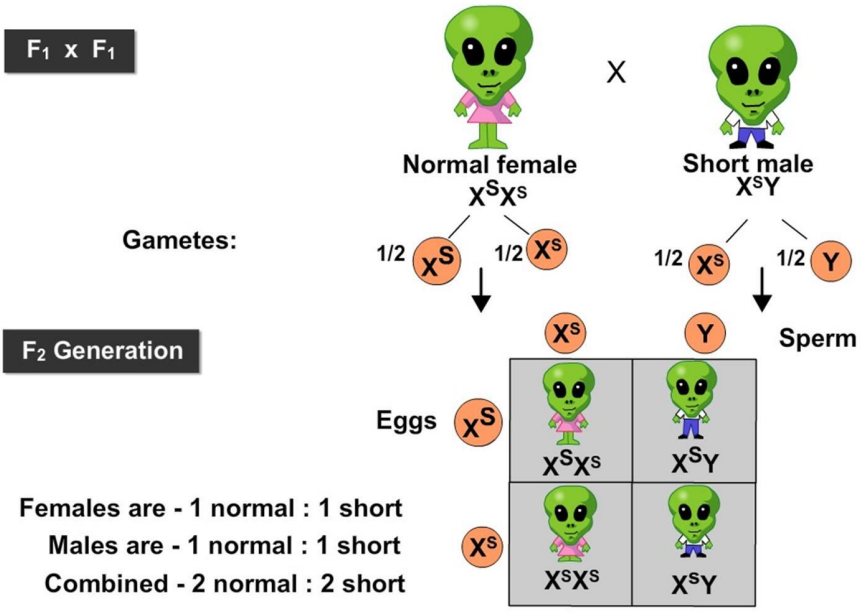


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If an X-linked trait is due to a recessive allele, a female will express the phenotype only if she is homozygous for that allele. Because males have only one locus, the terms *homozygous* and *heterozygous* lack meaning for describing their X-linked genes; the term *hemizygous* is used in such cases. Any male receiving the recessive allele from his mother will express the trait. For this reason, far more males than females have X-linked

recessive disorders. However, even though the chance of a female inheriting a double dose of the mutant allele is much less than the probability of a male inheriting a single dose, there *are* females with X-linked disorders. For instance, color blindness is almost always inherited as an X-linked trait. A color-blind daughter may be born to a color-blind father whose mate is a carrier (see Figure 15.7c). Because the X-linked allele for color blindness is relatively rare, though, the probability that such a male and female will mate is low.

A number of human X-linked disorders are much more serious than color blindness, such as **Duchenne muscular dystrophy**, which affects about one out of 5,000 males born in the United States. The disease is characterized by a progressive weakening of the muscles and loss of coordination. Affected individuals' life expectancy is into the mid-20s. Researchers have traced the disorder to the absence of a key muscle protein called dystrophin and have mapped the gene for this protein to a specific locus on the X chromosome. Since the gene is known, gene therapy is being explored.

**Hemophilia** is an X-linked recessive disorder defined by the absence of one or more of the proteins required for blood clotting. When a person with hemophilia is injured, bleeding is prolonged because a firm clot is slow to form. Small cuts in the skin are usually not a problem, but bleeding in the muscles or joints can be painful and can lead to serious damage. In the 1800s, hemophilia was widespread among the royal families of Europe. Queen Victoria of England is known to have passed the allele to several of her descendants. Subsequent intermarriage with royal family members of other nations, such as Spain and Russia, further spread this X-linked trait, and its incidence is well documented in royal pedigrees. A few years ago, new genomic techniques allowed sequencing of DNA from tiny amounts isolated from the buried remains of royal family members. The genetic basis of the mutation, and how it resulted in a nonfunctional blood-clotting factor, is now understood. Today, people with hemophilia are treated as needed with intravenous injections of the protein that is missing.

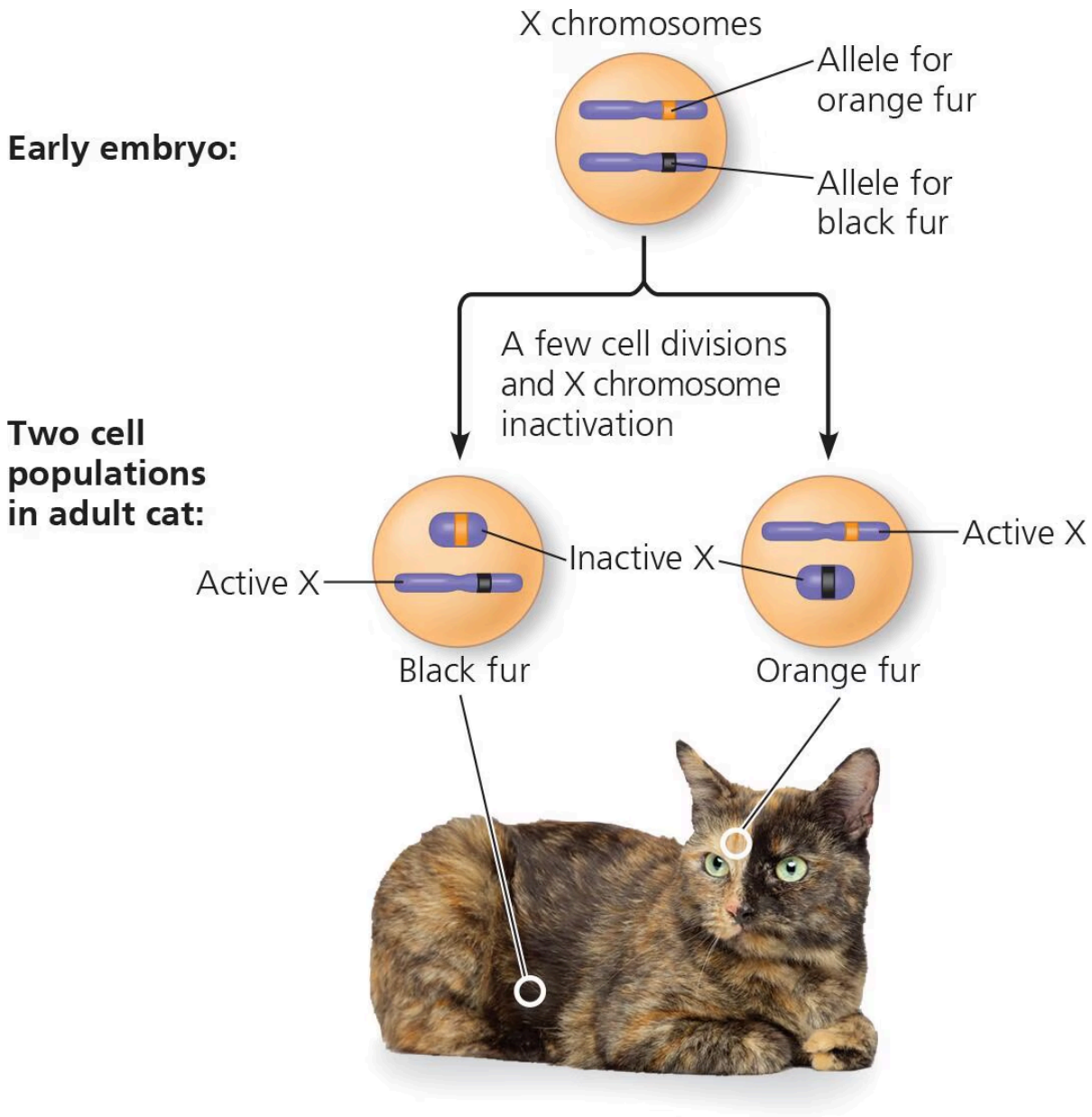
## X Inactivation in Female Mammals

Female mammals, including human females, inherit two X chromosomes—twice the number inherited by males—so you may wonder whether females make twice as many of the proteins encoded by X-linked genes as males. In fact, almost all of one X chromosome in each cell in female mammals becomes inactivated during early embryonic development. As a result, the cells of females and males have the same effective dose (one active copy) of most X-linked genes. The inactive X in each cell of a female condenses into a compact object called a **Barr body** (discovered by Canadian anatomist Murray Barr), which lies along the inside of the nuclear envelope. Most of the genes of the X chromosome that forms the Barr body are not expressed. In the ovaries, however, Barr body chromosomes are reactivated in the cells that give rise to eggs, resulting in every female gamete (egg) having an active X after meiosis.

British geneticist Mary Lyon demonstrated that the selection of which X chromosome will form the Barr body occurs randomly and independently in each embryonic cell present at the time of X inactivation. As a consequence, females consist of a *mosaic* of two types of cells: those with the active X derived from the male parent and those with the active X derived from the female parent. After an X chromosome is inactivated in a particular cell, all mitotic descendants of that cell have the same inactive X. Thus, if a female is heterozygous for a sex-linked trait, about half of her cells will express one allele, while the others will express the alternate allele. **Figure 15.8** shows how this mosaicism results in the patchy coloration of a tortoiseshell cat. In humans, mosaicism can be observed in a recessive X-linked mutation that prevents the development of sweat glands. A female who is heterozygous for this trait has patches of normal skin and patches of skin lacking sweat glands.

**Figure 15.8**

X inactivation and the tortoiseshell cat.



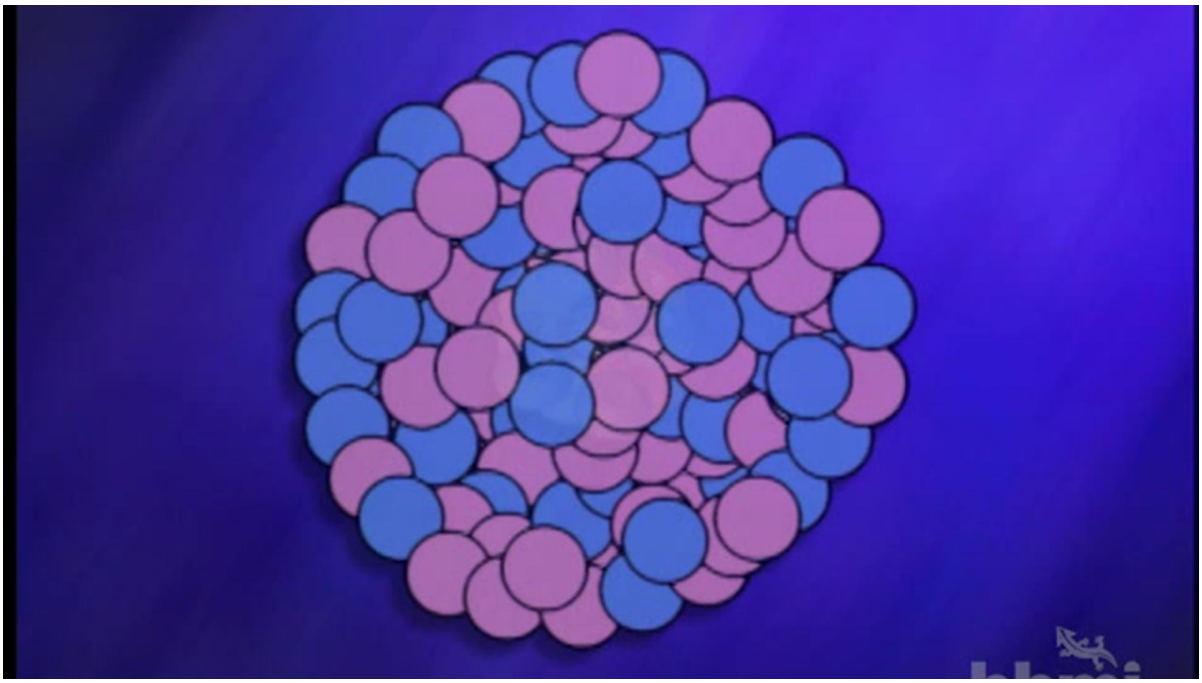

Early embryo is X X, one X with an allele for orange fur and one X with an allele for black fur. Following a few cell divisions and X chromosome inactivation, two cell populations exist in the adult cat: some cells with an active X chromosome with the allele for black fur and the other X chromosome inactivated (corresponds with black fur patches on a tortoiseshell cat), and some cells with an active X chromosome with the allele for orange fur and the other X chromosome inactivated (corresponds with patches of orange fur on a tortoiseshell cat).

The tortoiseshell gene is on the X chromosome, and the tortoiseshell phenotype requires the presence of two different alleles, one for orange fur and one for black fur. Normally, only females can have both alleles because only they have two X chromosomes. If a female cat is heterozygous for the tortoiseshell gene, she is

tortoiseshell. Orange patches are formed by populations of cells in which the X chromosome with the orange allele is active; black patches have cells in which the X chromosome with the black allele is active. (“Calico” cats also have white areas, which are determined by another gene.)

**HHMI BioInteractive Animation**

X Inactivation



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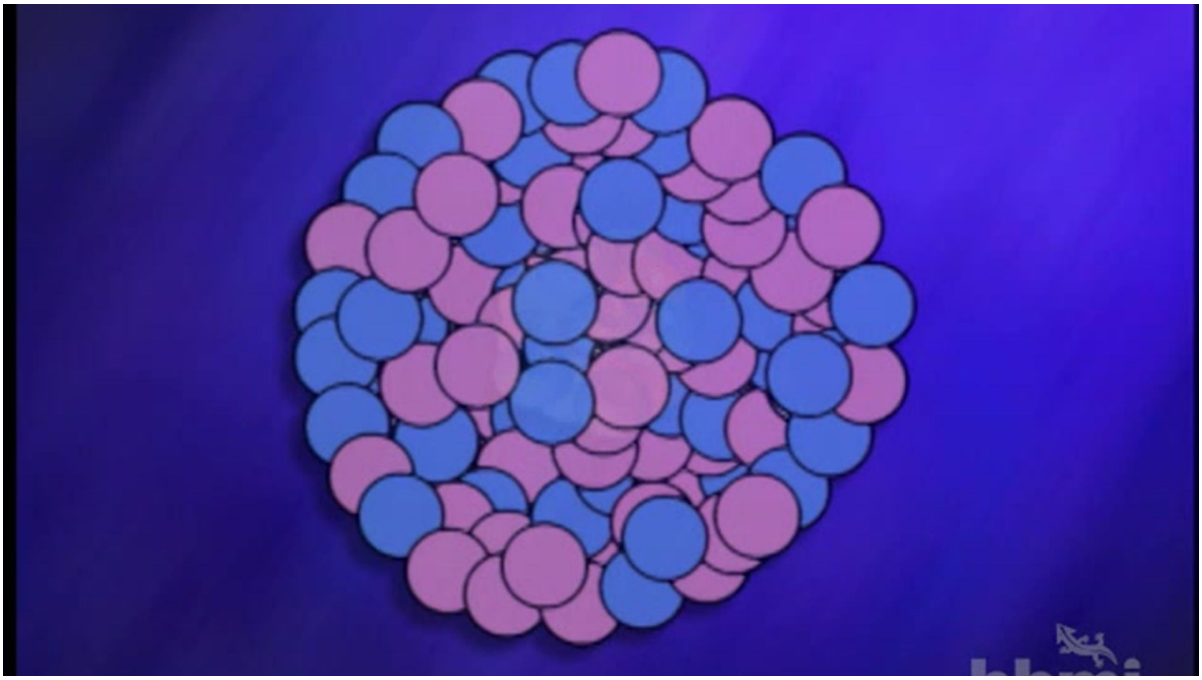

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Inactivation of an X chromosome involves modification of the DNA and proteins bound to it called histones, including attachment of methyl groups (—$CH_3$) to DNA nucleotides. (The regulatory role of DNA methylation is discussed in Concept 18.2.) A particular region of each X chromosome contains several genes involved in the inactivation process. The two regions, one on each X chromosome, associate briefly with each other in each cell at an early stage of embryonic development. Then one of the genes, called *XIST* (for X-inactive specific transcript), becomes active *only* on the chromosome that will become the Barr body (the inactive X). Multiple copies of the RNA product of this gene apparently attach to the X chromosome on which they are made, eventually almost covering it. Interaction of this RNA with the chromosome initiates X inactivation, and the RNA products of nearby genes help to regulate the process.

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